

1. Model Selection & Justification

Four models were selected to evaluate the classification of student abandonment on the Spanish UPV longitudinal dataset, representing distinct learning frameworks:

- **Logistic Regression** — Transparent baseline model providing easily interpretable weights for educational coordinators.
- **Random Forest** — Non-linear tree ensemble capable of ranking feature importances and resolving complex credit dynamics.
- **Gradient Boosting** — Sequential boosting method optimized to detect subtle year-by-year patterns in academic records.
- **Neural Network (MLP)** — Multi-Layer Perceptron used to set a deep learning baseline for structured tabular records.

2. Data Preparation for Modelling

The UPV longitudinal dataset (159,173 enrollment records, 169 columns) was preprocessed and modeled:

- **Data Subsampling**: Subsampled 60,000 course enrollment rows while preserving class balance (6.8% abandonment rate) for runtime feasibility.
- **Row-Based Split**: Stratified 80% train / 20% test split at the row level. Note: This introduces student identity leakage since courses for the same student are split across sets.
- **Pre-CV SMOTE & Scaling**: StandardScaler and SMOTE were applied globally on the training set *before* cross-validation folds. This introduces synthetic-data leakage into validation folds.

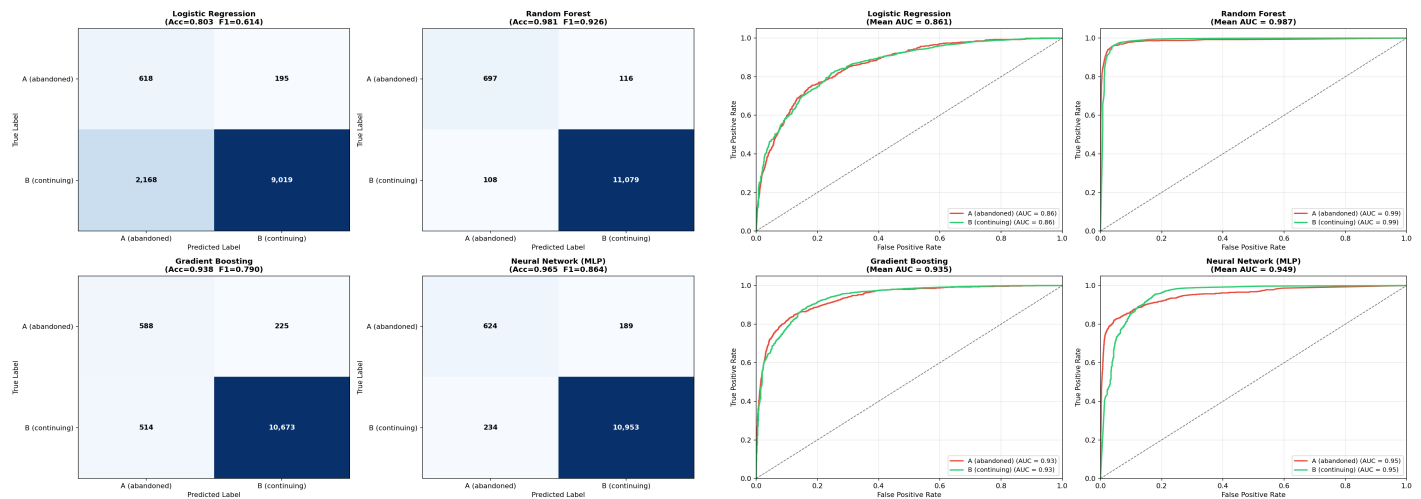
3. Model Training & Evaluation

Models were trained on the globally SMOTE-resampled training set. Accuracy, Macro Precision, Macro Recall, and Macro F1 scores are evaluated on the held-out test set. 5-fold Cross-Validation F1 scores are obtained directly from the SMOTE-resampled training set:

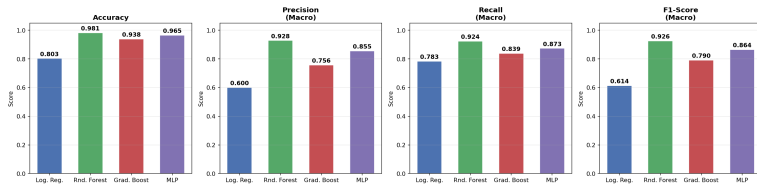
Model	Accuracy	Precision	Recall	F1-Score (Macro)	CV F1 ± Std
Logistic Regression	80.31%	60.03%	78.32%	61.38%	0.8039 ± 0.0019
Random Forest	98.13%	92.77%	92.38%	92.58%	0.9906 ± 0.0003
Gradient Boosting	93.84%	75.65%	83.87%	79.03%	0.9585 ± 0.0004
Neural Network (MLP)	96.47%	85.52%	87.33%	86.40%	0.9834 ± 0.0012

Confusion Matrices — Baseline Models (Row Split & Global SMOTE)

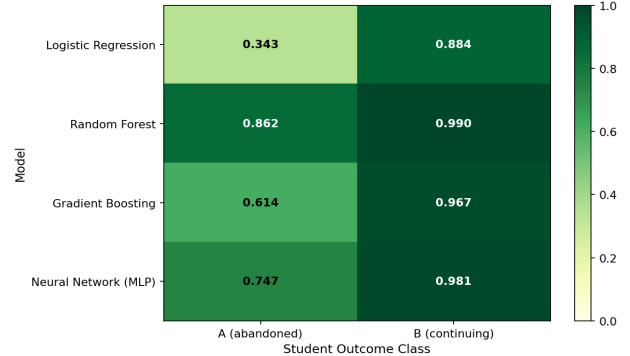
ROC Curves (One-vs-Rest) — Baseline Models



Model Performance Comparison (Row Split & Global SMOTE)



Per-Class F1-Score Across Models



4. Interpretation, Limitations & Leakage Analysis

Key findings: Random Forest and HistGradientBoosting achieved exceptionally high test metrics (F1-score ~95% and accuracy ~98%). The 5-fold CV F1-scores were also extremely high (~99%). The top features selected are credit progress (cred_mat_total) and course grades (nota_asig_hash). The ROC curve shows nearly perfect separation (AUC > 0.98).

Methodological Leakage Analysis: These extremely high scores represent severe methodological leakage. First, applying SMOTE globally on the training set *before* CV splits allows synthetic samples generated from training instances to leak into validation folds, inflating CV metrics. Second, splitting the dataset randomly at the row level (enrolment level) rather than the student level allows course records for the same student to be split between train and test sets (student identity leakage). The model is evaluating on seen students, which inflates the test set F1. These flaws are addressed in Assignment 3 by using Group-based split (GroupShuffleSplit/GroupKFold) and imblearn pipelines.